ASSIGNMENT 003: LEARNING REPORT

PHP & MySQL Dynamic Website and Admin Dashboard

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1. Introduction and Objectives

Assignment 003 extends the static Nexora Digital Solutions website from Assignment 002 by introducing server-side PHP functionality and a MySQL database design. The main aim is to make website content data-driven and provide an administration area for managing services, portfolio projects, and visitor enquiries. The work also focuses on secure coding practices, clear project documentation, and version control using GitHub.

The objectives were to connect PHP to MySQL using PDO, display services and projects dynamically, implement administrator login and logout, provide create/read/update/delete (CRUD) operations, store contact enquiries, and validate input on the server. Additional objectives included documenting the database structure, providing setup instructions, and preparing a testing report.

2. Work Completed

A project structure was organised under **assignments/assignment-003/**. It includes public website pages, shared header and footer files, a configuration directory, database documentation, and an admin directory. The shared layout files help keep the public pages consistent, while the stylesheet provides common presentation.

The database design documents four tables: **admin_users**, **services**, **projects**, and **enquiries**. The **database/schema.sql** file describes the tables, **database/ERD.md** documents the entity relationships, and **database/sql-queries.md** records SQL examples. A **.env.example** file lists the database configuration values required locally without including real credentials.

The public pages include the home page, services page, portfolio page, and contact form. The services and portfolio pages use database queries to retrieve content. The contact form validates submitted values, checks a CSRF token, uses a prepared statement to insert enquiries, and escapes output displayed back to the visitor.

The admin area includes login and logout, a dashboard, service and project management pages, and enquiry listing/status management. The reviewed code uses password hashing and verification, prepared statements, session checks, CSRF tokens on relevant forms, and output escaping. The initial administrator setup page is intended only for trusted local development and must be removed or disabled before public deployment.

3. Security and Documentation

Security was treated as an important part of the assignment. Database queries use PDO prepared statements to reduce SQL injection risk. Passwords are stored as hashes rather than plain text. Server-side validation checks submitted data, and output is escaped to reduce the risk of cross-site scripting. Session ID regeneration after successful login helps protect authenticated sessions, while CSRF tokens are used to help prevent unauthorised form submissions.

Session cookie settings were strengthened in the administrator login and contact form. The setup page was also updated to check a CSRF token and apply input limits. These changes have been reviewed in the source code, but they have not yet been executed in a live PHP/MySQL environment. They therefore still require runtime testing and further security review before any production use.

The README explains the project objectives, technologies, database, security requirements, local setup approach, and testing status. It also notes that PHP does not automatically load values from **.env.example**; the environment variables must be configured in the local server environment. The testing report records planned test cases, and test statuses should only be changed after each test has actually been run.

4. Challenges and Lessons Learned

One challenge was organising files into the correct GitHub directory and checking that names and paths matched the code. Working through this helped reinforce the importance of consistent filenames, relative paths, and a clear folder structure. Another challenge was preparing a PHP/MySQL environment on the available computer. Because the local PHP setup was not completed, the project has been maintained and reviewed through GitHub, but it has not been fully run locally.

The assignment reinforced the value of small, focused Git commits, careful review of code changes, and accurate documentation. It also demonstrated that a website can appear complete in source files while still needing execution tests to confirm database connections, form submissions, authentication, and CRUD operations work as expected. Security features must be reviewed together and verified through testing rather than assumed to work solely because code has been written.

5. Testing Status and Future Improvements

At the time of writing, runtime tests are **not yet run**. The source code has been reviewed during development, but this is not the same as successful execution. The next step is to configure PHP with PDO MySQL and a MySQL server, import **database/schema.sql**, configure the required environment variables, and run each case in **testing-report.md**. Results should be recorded as passed or failed only after observing the actual outcome. Screenshots should also be captured from the running public website and admin dashboard when the environment is available.

Future improvements include completing runtime and security testing, confirming all file paths, removing or disabling the temporary administrator setup page, and adding any required screenshots and final evidence. The project should not be deployed publicly until configuration, authentication, access control, form handling, and database operations have been tested and reviewed.

6. Conclusion

Assignment 003 has provided practical experience in PHP, MySQL database design, PDO, dynamic content, administrator authentication, CRUD workflows, form validation, CSRF protection, documentation, and GitHub-based version control. The core project files and documentation have been organised, and several security improvements have been added. Completing the local setup, running the test cases, recording evidence, and finalising screenshots remain necessary before the implementation can be described as fully tested.